

OPERATOR'S GUIDE



USA Instruments, Inc.

Document 2377426-4

Revision 3

**GE Signa® 1.5T
PREMIER 9000
8-CHANNEL CTL COIL**

GE Catalog Part Number: M3087JC

USAI Part Number: U1-770015

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SECTION 1 – INTRODUCTION

1-1 Operator's Guide Information

The operator's guide provides information on how to use the 8-Channel CTL Coil. The user should read this guide before using the coil for imaging. The user is advised to refer to the MR scanner manual in conjunction with this guide.

Direct any questions regarding this manual to:

USA Instruments, Inc.
1515 Danner Drive
Aurora, Ohio USA 44202

1-2 Use of Symbols

Coil users should refer to the operator's guide for caution and warning explanations, using the symbols defined below.



Cautions provide information on possible equipment hazards.



Warnings provide information on possible hazards to human beings.

1-3 Coil Classification

Class II equipment (the symbol is shown right).



Type BF applied part (the symbol is shown right).



Ordinary equipment.

Suitable for continuous operation.

1-4 Safety

1-4-1 Coil Operating Safety

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may potentially cause injury.

1-4-2 Patient Safety



Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.

1-4-3 Equipment Safety

All personnel using the coil should be instructed in the proper connection and handling of the coil.

Personnel should observe all warnings and cautions that appear in this guide.



Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.



The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.



Prevent cables from forming loops. Looping will degrade scan performance of the coil through RF coupling.

1-4-4 Electrical and Mechanical Safety



The coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it to ensure there is no external damage. Do not use the coil if the housing or cable is broken.



WARNING: This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



1-4-5 Modifications

Any changes or modifications to the coil must be approved and performed by USA Instruments prior to installation.

1-4-6 Reporting of Incidents

Users should contact USA Instruments immediately to report an incident and/or injury to a patient, operator or maintenance employee that occurred as a result of coil operation. If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

1-5 Coil Description

The 8-Channel CTL Coil is a receive-only coil designed to give optimum signal to noise ratio and uniform coverage of the spine anatomy, including the cervical, thoracic and lumbar regions. The coil has a removable cervical bridge for enhanced imaging of the anterior cervical region.

The coil allows the user to obtain high-resolution images of extended regions (such as the cervical-thoracic spine and thoracic-lumbar spine) or specific regions of interest (such as the cervical spine). The coil has different modes of operation, including: (1) cervical CTL mode for cervical imaging, (2) thoracic CTL mode for thoracic imaging, and (3) lumbar CTL mode for lumbar imaging.

Each coil section includes two elements and uses two channels through the interface (such as 8CTL345 uses three channels and six elements). The modes are enabled by the clinical user at the operator console. Other modes and configurations may be provided by GE Medical Systems.



United States federal law restricts this device to sale by, or on the order of, a physician.



Cables bearing this Coil ID label are compatible with software versions 10x and newer. With software versions 11x and newer, the system will generate an error for an incompatible coil.

1-6 Coil Mode Configurations

The main goal of this document is to help the user optimize image quality using the 8-Channel CTL Array, focusing on the coil configuration, patient positioning and FOV selection. Improper use of the coil is the major cause of image artifacts.

The coil has a total of six sections which are activated according to the coil configuration file selected (see *Figure 1*).

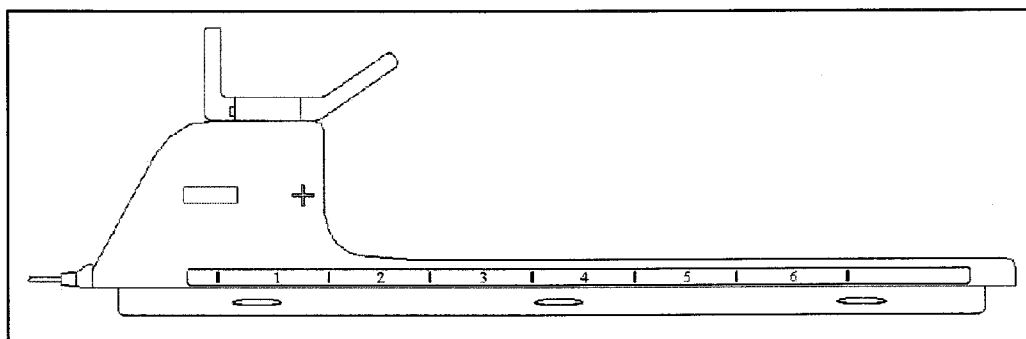


Figure 1: The 8-Channel CTL Array with alignment marks.

There are 12 coil configuration modes from which to choose:

8CTLTOP: Eight coil elements are active for this coil configuration (see *Figure 2*).

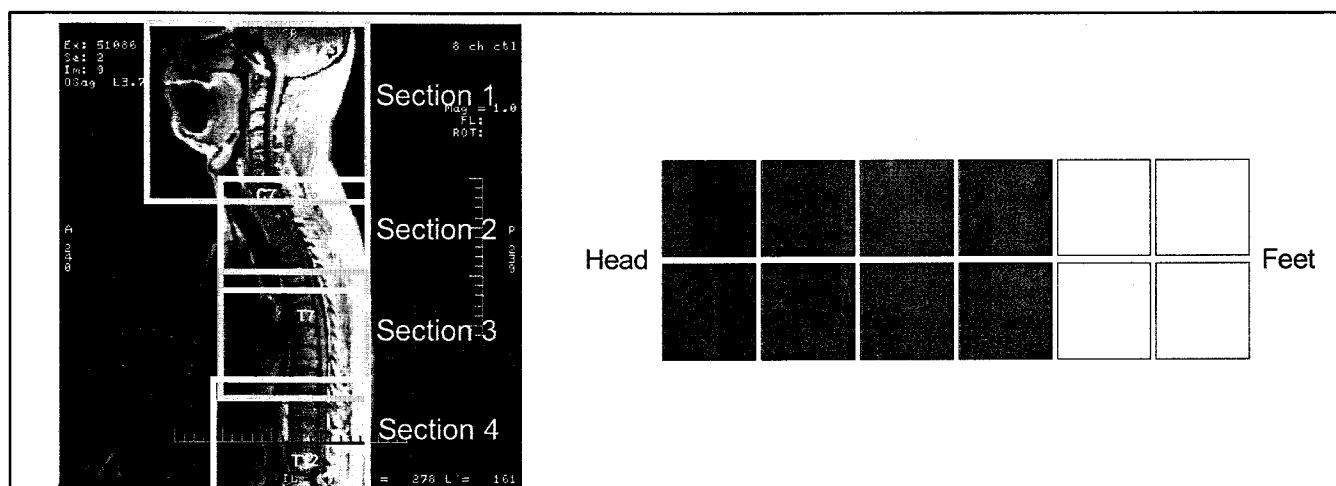


Figure 2: 8CTLTOP active sections.

8CTLMID: Eight coil elements are active for this coil configuration (see *Figure 3*).

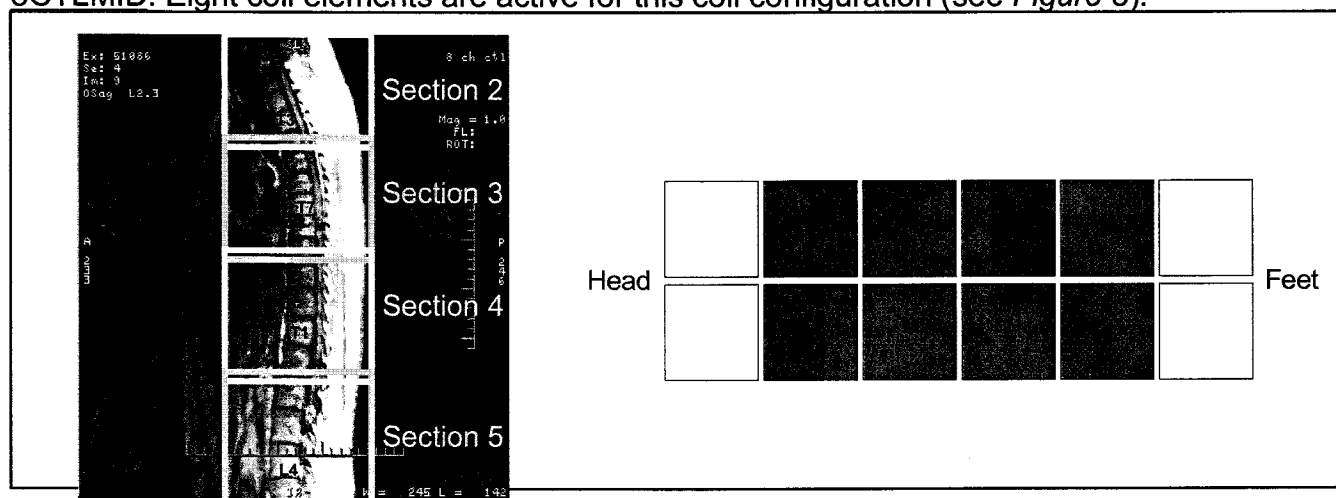


Figure 3: 8CTLMID active sections.

8CTLBOT: Eight coil elements are active for this coil configuration (see *Figure 4*).

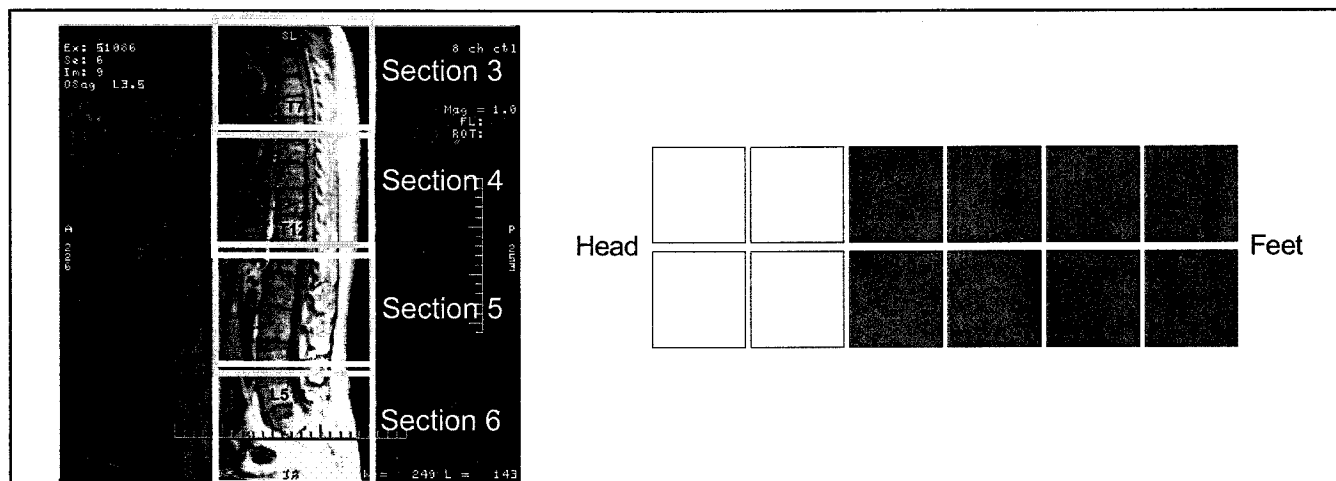


Figure 4: 8CTLBOT active sections.

8CTL123: Six coil elements are active for this coil configuration (see *Figure 5*).

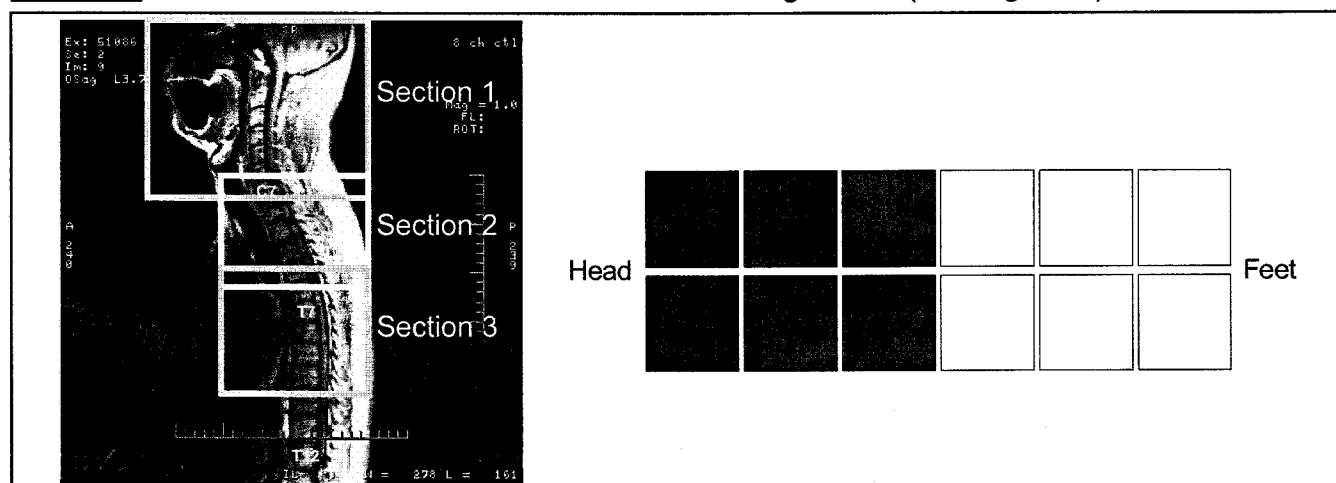


Figure 5: 8CTL123 active sections.

8CTL234: Six coil elements are active for this coil configuration (see *Figure 6*).

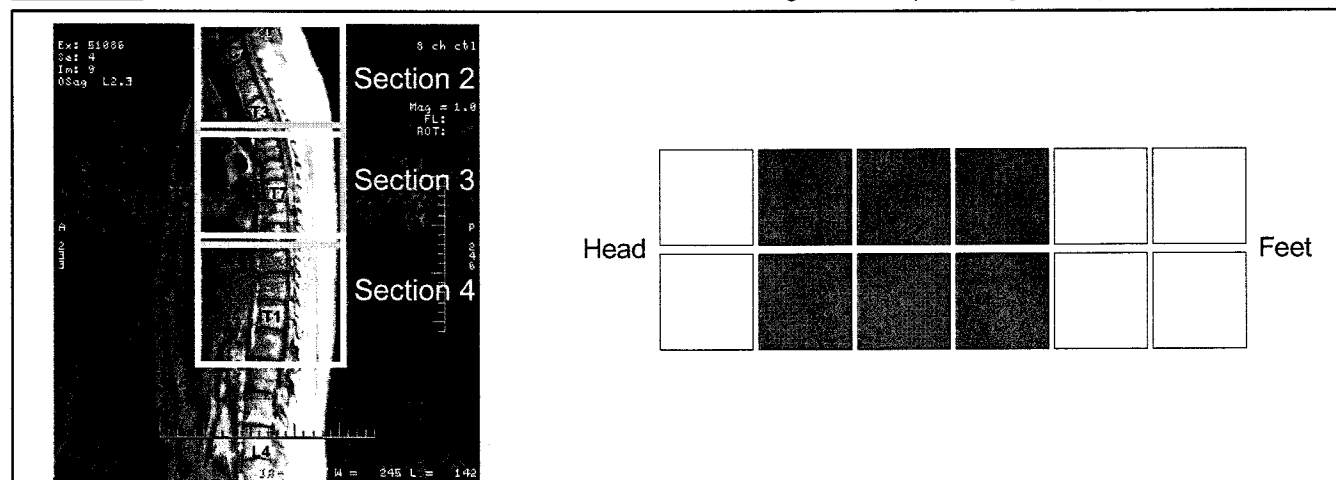


Figure 6: 8CTL234 active sections.

8CTL345: Six coil elements are active for this coil configuration (see *Figure 7*).

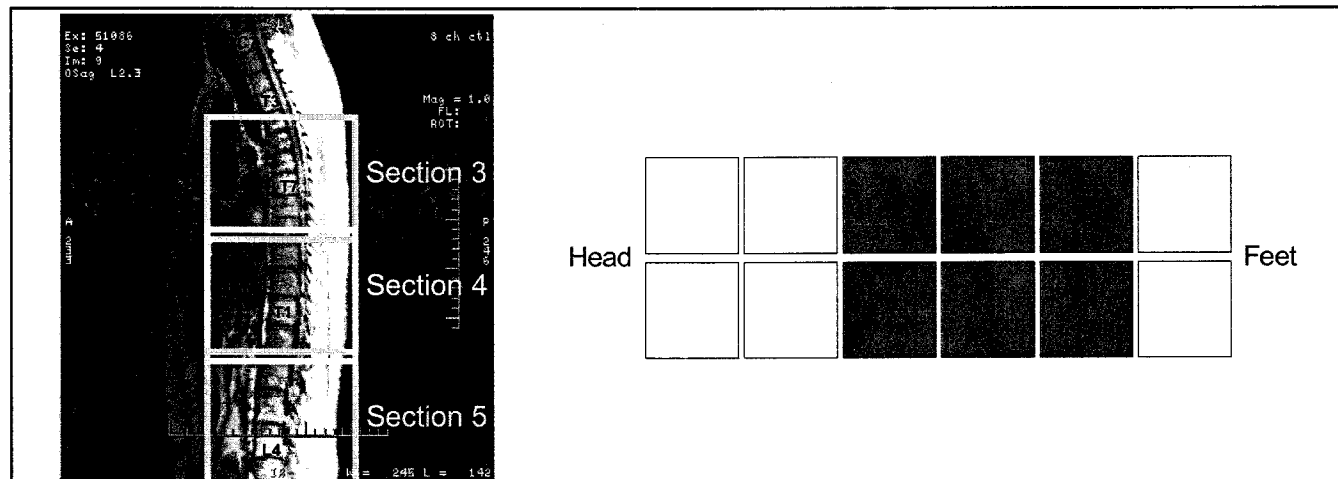


Figure 7: 8CTL345 active sections.

8CTL456: Six coil elements are active for this coil configuration (see *Figure 8*).

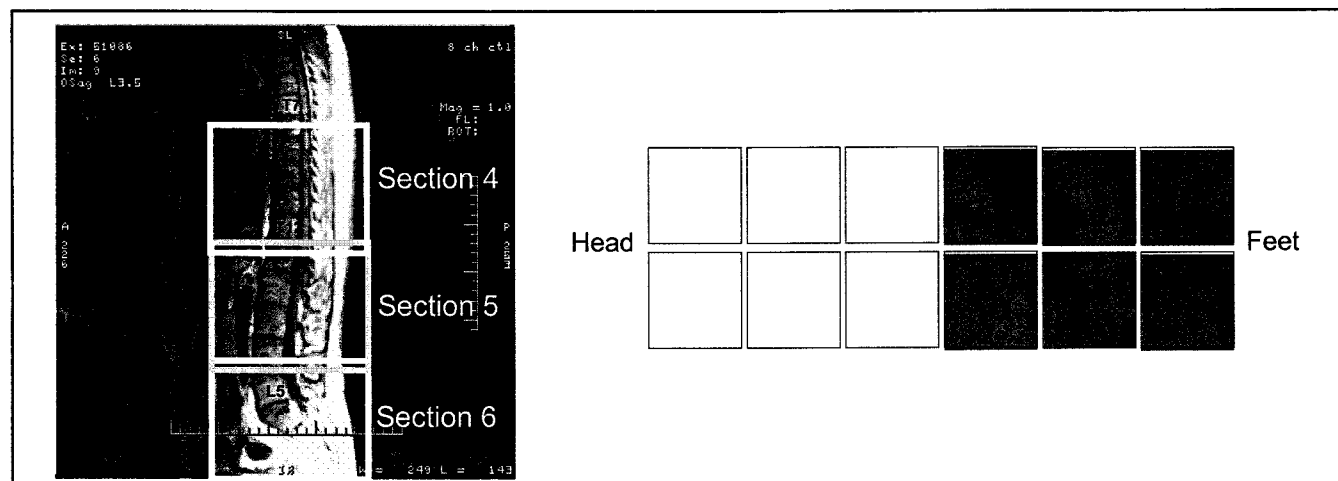


Figure 8: 8CTL456 active sections.

8CTL12: Four coil elements are active for this coil configuration (see *Figure 9*).

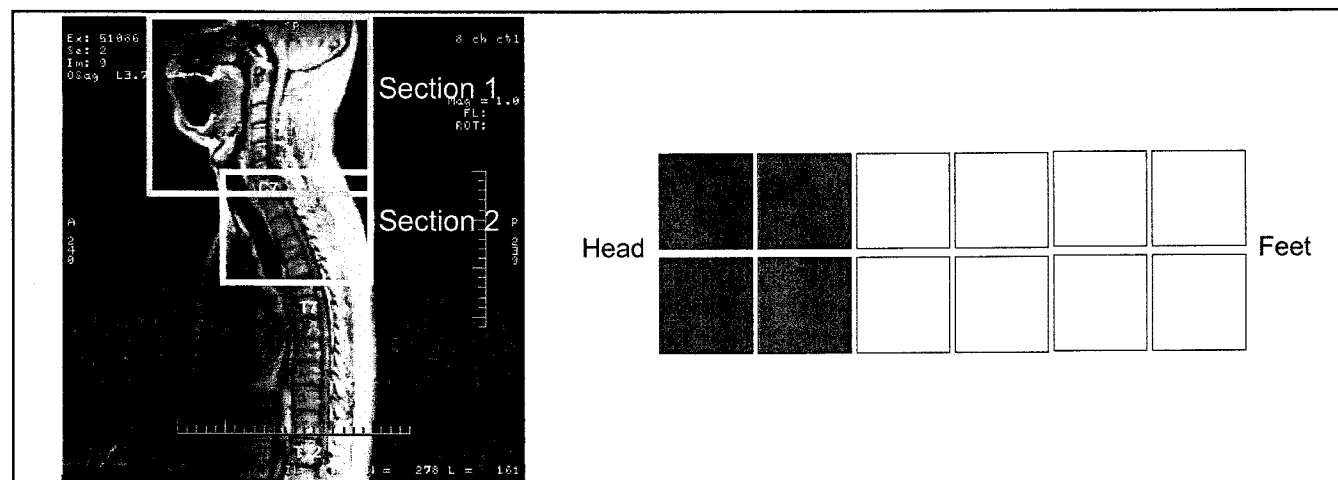


Figure 9: 8CTL12 active sections.

8CTL23: Four coil elements are active for this coil configuration (see *Figure 10*).

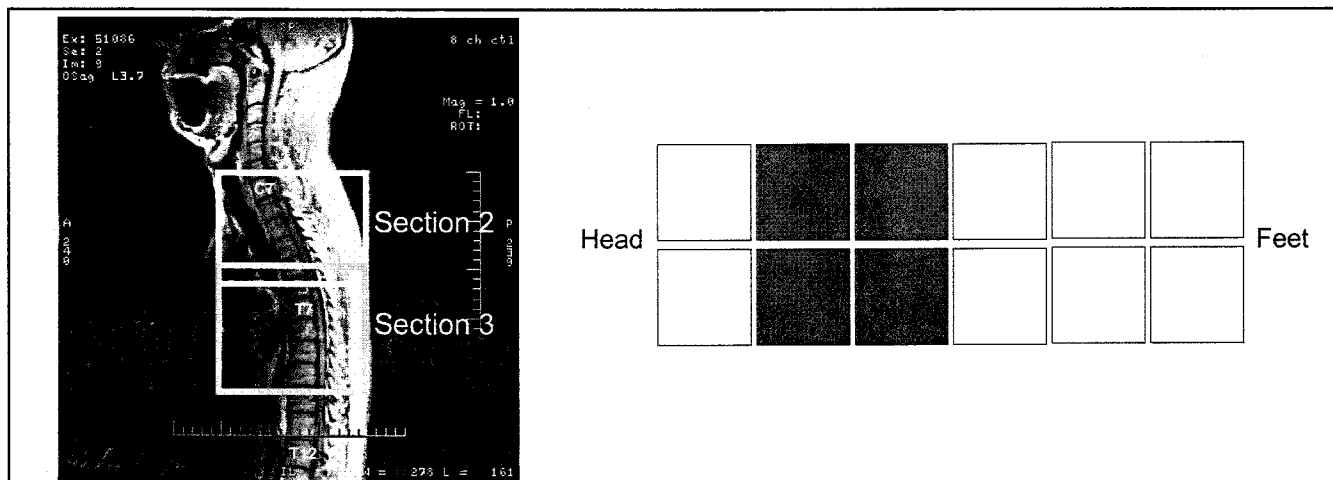


Figure 10: 8CTL23 active sections.

8CTL34: Four coil elements are active for this coil configuration (see *Figure 11*).

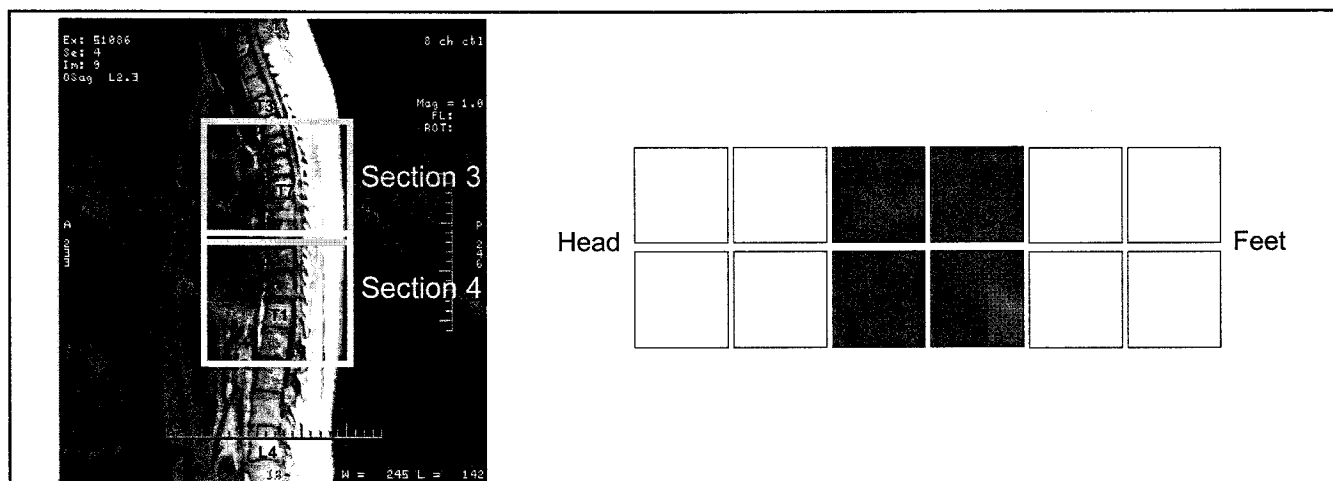


Figure 11: 8CTL34 active sections.

8CTL45: Four coil elements are active for this coil configuration (see *Figure 12*).

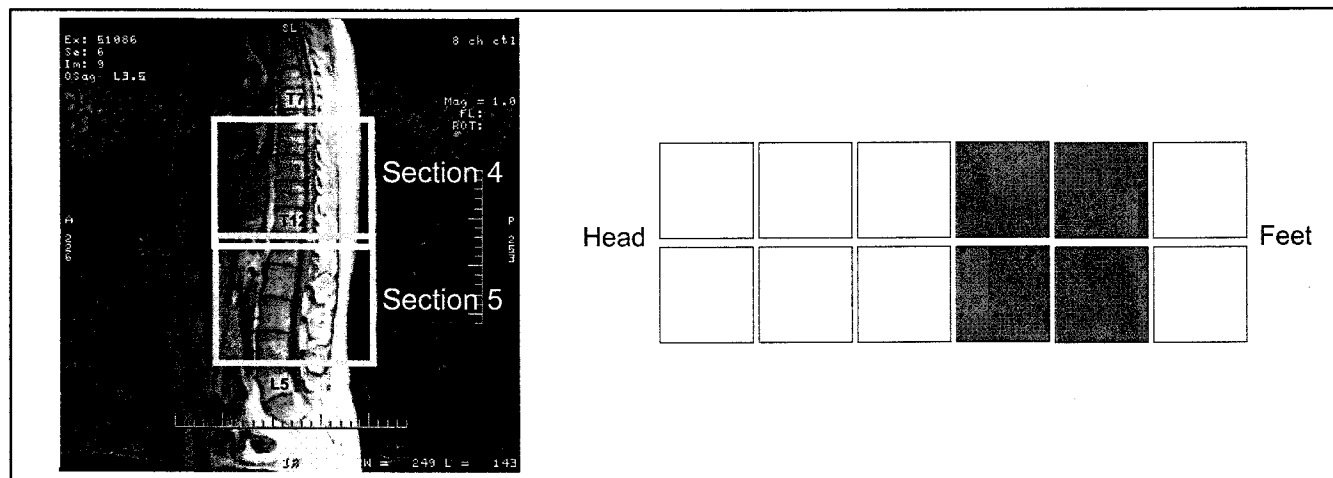


Figure 12: 8CTL45 active sections.

8CTL56: Four coil elements are active for this coil configuration (see *Figure 13*).

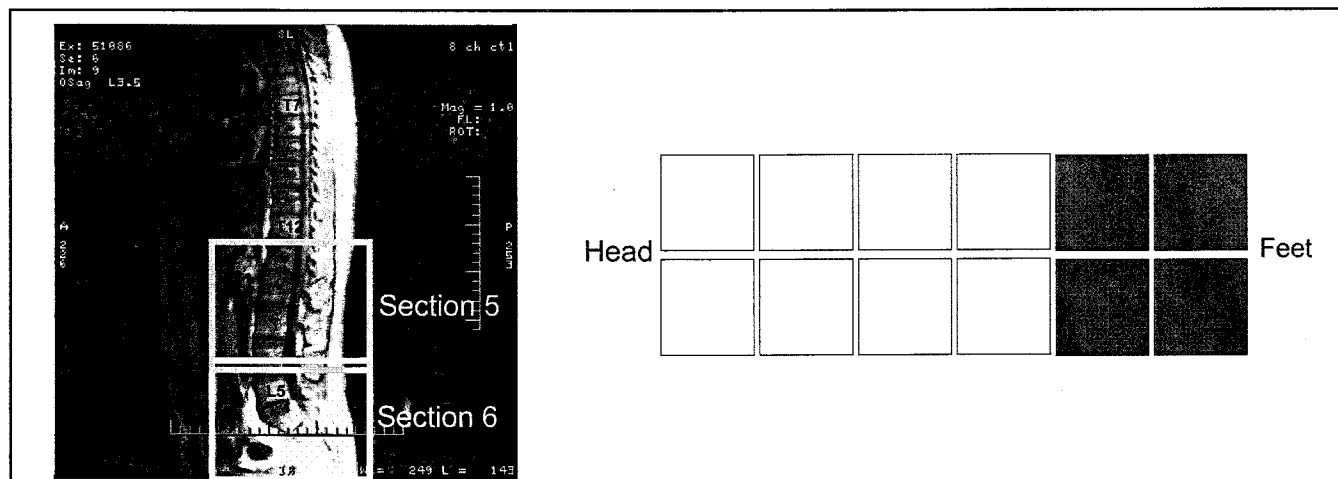


Figure 13: 8CTL56 active sections.

1-7 Environmental Requirements

ENVIRONMENTAL REQUIREMENTS – TABLE 1-7

	Transport and Storage	Operating Conditions
Barometric Pressure	645mm Hg to 795mm Hg	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	15° C to 35° C (59° F to 95° F)

Dimensions

43.3" L x 16.1" W x 13.6" H (110.0 cm x 41.0 cm x 34.5 cm)

Weight

18 lbs. (8.1 kg)

1-8 Coil Specifications

COIL SPECIFICATIONS – TABLE 1-8

Field Strength	Coil Model	USAI Part Number	Frequency
1.5T	Premier 9000	100341	63.86 MHz

1-9 Coil Installation and Service

1-9-1 Coil Installation Procedure

Coil installation requires the services of your GE Service Representative.

1-9-2 Coil Service

The coil cannot be serviced on site. Contact your authorized GE Service Representative for service and maintenance needs.

SECTION 2 – SETUP

2-1 Positioning the Coil



WARNING!

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Step 1 – Position the coil on the cradle near the bore of the magnet (see *Figure 14*).

Step 2 – Plug the coil connector into the RF coil receptacle. The coil connector and cable should always be routed directly to the magnet and should not form loops.



Figure 14: Positioning the coil on the scanner cradle.

2-2 Positioning the Patient

2-2-1 Head First Imaging

Step 1 – Remove the cervical bridge of the coil.



CAUTION

The proper way to remove the cervical bridge is to press both release buttons simultaneously and lift the bridge using both hands. Pressing only one button and lifting the bridge will damage the coil (see *Figure 15*).

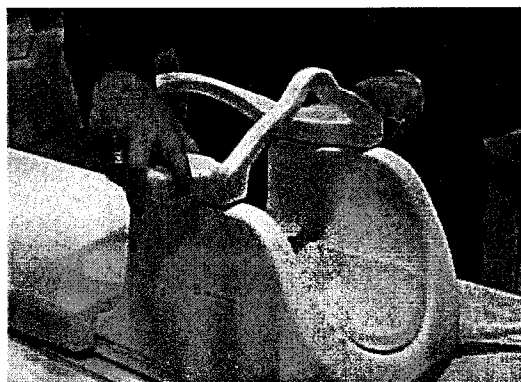


Figure 15: Removing the cervical bridge.



The coil is supplied with an adapter block that should be used when the cervical bridge is not used during scanning. The adapter block will complete the circuit for optimal imaging.

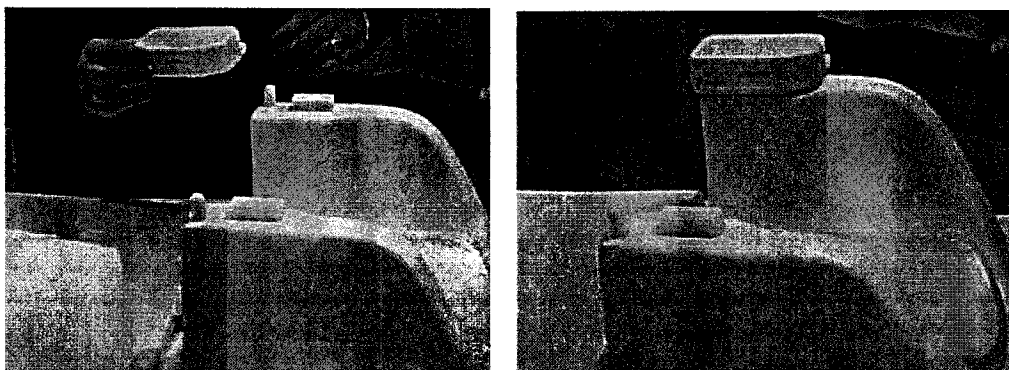


Figure 16: Attaching the adapter block.

Step 2 – Place a clean cotton sheet over the coil so that the patient's skin will not come into direct contact with the coil.

Step 3 – Position the patient head first, supine, with the top of the shoulders snugly resting against the coil sides.

Step 4 – After the patient has been positioned as described in Step 3, place the cervical bridge directly onto the posterior section by:

- (1) Pressing the button in on either side of the bridge,
- (2) Pressing the bridge onto the posterior section (alignment marks are available to facilitate proper alignment), and
- (3) Gently releasing the buttons.



Figure 17: Positioning the patient head first on the coil.



The cervical bridge should remain attached to the posterior section of the coil when scanning the cervical and upper thoracic spine. When imaging the lower thoracic and lumbar spine, the adapter block should be used in place of the cervical bridge.

2-2-2 Feet First Imaging

Step 1 – Remove the cervical bridge of the coil and attach the adapter block (see *Figures 15 and 16*).

Step 2 – Position the knee transition pad over the cervical end of the coil.

Step 3 – Place a clean cotton sheet over the coil so that the patient's skin will not come into direct contact with the coil.

Step 4 – Position the patient feet first, placing them snugly against the knee transition pad (see *Figure 18*).



Figure 18: Positioning the patient feet first on the coil.

2-3 Landmarking the Anatomy

Position the alignment light on the anatomy of interest and landmark (see *Figure 19*).

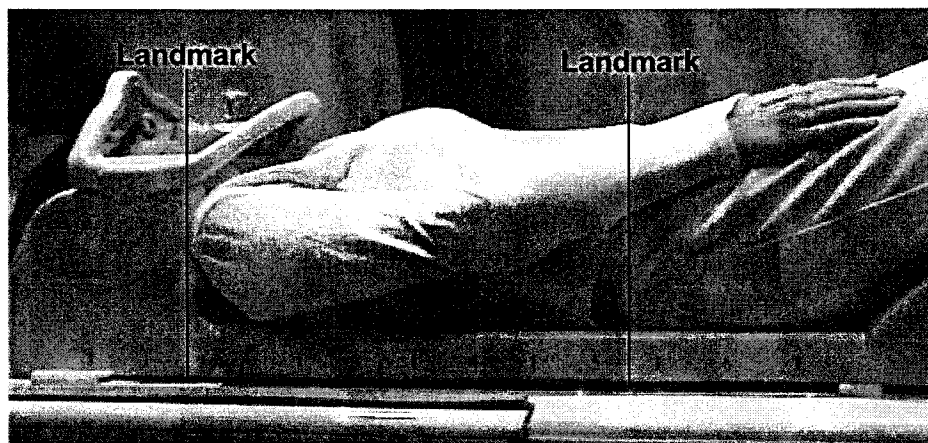


Figure 19: Landmarking for cervical or lumbar scanning (exact locations may change for some patients).

2-4 Choosing the Correct Coil Mode Configuration

Refer to Section 1-6 for detailed descriptions of each coil mode configuration. Choose the configuration most appropriate for the corresponding anatomy of interest.

The 8-Channel CTL Array is provided with external marks that represent each coil element.

Step 1 – Landmark on the coil marker, not on the anatomy.

Step 2 – Select the appropriate number of elements according to the area that needs to be covered.

Step 3 – Landmark on an imaginary middle line, between the elements used.

Step 4 – Considering that each element is 12cm, select a coil configuration and FOV based on the number of elements chosen.

To reduce artifacts in sagittal or coronal imaging, use the minimum number of coil sections to image the given FOV. For any 4-section coil selection, the maximum FOV is 48cm. For any 3-section selection, the maximum FOV is 36cm. For any 2-section selection, the maximum FOV is 25cm.

2-5 Setting up the Scan

Note: Auto-N-Coil is only compatible with 4-Channel CTL coils and will disable elements in the middle of a given series producing undesired image artifacts.

The coil is designed to be compatible with recommended protocols suggested by the system's manufacturer. Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

The procedure described in section 2-4 is seen as the best practice for the coil usage. If you are still unsure of coil configuration selection for the region of interest the control variable APA will determine the best choice for FSE sequences only.

Step 1 – Start with a coil selection of either: **8CTLTOP**, **8CTLMID** or **8CTLBOT**.

Step 2 – Enable the APA feature under the User CV page.

Step 3 – Based on the landmark and FOV, the scanner will determine which coils should be used for optimal image quality.

The image annotation does not record the selected coils by APA.

Cervical Spine 8CTL12	SCAN 1	SCAN 2	SCAN 3	SCAN 4	SCAN 5	SCAN 6	SCAN 7	SCAN 8	SCAN 9	SCAN 10
Description	SagT1SE	SagT1FSE	SagPDFSE	SagT2FSE	AxT2GRE	AxT2FGRE	AxT1SE	SagT1Flair	SagT2FRFSE	AxT2FRFSE
Coil	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12	8CTL12
Plane	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique
Mode	2D	2D	2D	2D	2D	2D	2D	2D	2D	2D
Pulse Sequence	SE	FSE-XL	FSE-XL	FSE-XL	GRE	FGRE	SE	T1Flair	FRFSE-XL	FRFSE-XL
Imaging Options	VB	TRF	NPW TRF	NPW	FC VB	FC VB	VB	NPW SEQ VB TRF ZIP512	NPW TRF ZIP512	FC NPW TRF ZIP512
PSD Filename										
No. Echos/Shots	1	1	1	1	1	1	1	1	1	1
TE 1/TE 2	MinFull	MinFull	MinFull	102	17	MinFull	MinFull	24	110	110
TR	450	550	2000	3500	625	550	650	2000	3300	2500
Inv. Time								860		
Flip Angle					20	20				
Echo Train Length		2	6	16				6	33	29
RBW 1/ RBW 2	15.63	20.83	20.83	20.83	10.42	6.94	10.42	20.83	41.67	31.25
FOV	24	24	24	24	22	22	20	24	24	20
Slice Thickness	3	3	3	3	4	4	4	3	3	4
Slice Spacing	1	1	1	1	0	0	0	1	1	0
No. Scan Locs										
No. Slices/Slabs										
Frequency	256	256	256	256	256	256	256	256	256	256
Phase	224	224	224	192	192	192	192	224	224	224
NEX	2	3	2	3	3	3	3	3	4	4
Phase FOV	1	1	1	1	0.75	0.75	1	1	1	1
Freq Direction	S/I	S/I	A/P	A/P	R/L	R/L	R/L	A/P	A/P	R/L
Flow Comp Direction										
Autoshim On?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Phase Correct On?	no	yes	yes	no	no	no	no	no	no	no
No. Locs B4 Pause/Phase										
Sat Details	Anterior	Anterior	Anterior	Anterior	Anterior	Anterior	Inf / Sup	Inf /Sup	Anterior	Anterior
SCIC On?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
User CVs Details										
Vascular Screen Info										
Triggering Details										
Cine Details										
Multi-Phase Details										
Diffusion Screen										
Other Relevant Info:										

Thoracic 8CTL234	SCAN 1	SCAN 2	SCAN 3	SCAN 4	SCAN 5	SCAN 6	SCAN 7	SCAN 8	SCAN 9
Description	SagT1SE	SagT1FSE	SagT2FSE	AxT2GRE	AxT2FSE	AxT1SE	SagT1Flair	SagT2FRFSE	AxT2FRFSE
Coil	8CTL234	8CTL234	8CTL234	8CTL234	8CTL234	8CTL234	8CTL234	8CTL234	8CTL234
Plane	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique
Mode	2D	2D	2D	2D	2D	2D	2D	2D	2D
Pulse Sequence	SE	FSE-XL	FSE-XL	GRE	FSE-XL	SE	T1Flair	FRFSE-XL	FRFSE-XL
Imaging Options	VB NPW	NPW TRF	NPW	FC VB NPW	FC NPW VB	VB NPW	NPW SEQ VB TRF ZIP512	NPW TRF ZIP512 VB	FC NPW TRF ZIP512 VB
PSD Filename									
No. Echos/Shots	1	1	1	1	1	1	1	1	1
TE 1/TE 2	MinFull	MinFull	102	17	102	MinFull	24	110	110
TR	450	450	3500	550	5000	550	1800	3200	3300
Inv. Time							650		
Flip Angle				20					
Echo Train Length		2	16		14		6	27	29
RBW 1/ RBW 2	15.63	20.83	20.83	12.5	20.83	15.63	31.25	41.67	31.25
FOV	34	34	34	20	20	20	34	34	20
Slice Thickness	3	3	3	4	4	4	4	3	4
Slice Spacing	1	1	1	1	1	1	0	1	1
No. Scan Locs									
No. Slices/Slabs									
Frequency	256	256	256	256	256	256	256	448	256
Phase	256	256	256	192	224	192	256	224	192
NEX	2	3	4	3	4	3	4	4	4
Phase FOV									
Freq Direction	A/P	A/P	A/P	A/P	A/P	A/P	A/P	A/P	A/P
Flow Comp Direction									
Autoshim On?	yes	yes	yes	yes	yes	yes	yes	yes	yes
Phase Correct On?	no	no	no	no	no	no	no	no	no
No. Locs B4 Pause/Phase									
Sat Details	Anterior	Anterior	Anterior		Sup / Inf		Sup / Inf	Anterior	Sup / Inf
SCIC On?	yes	yes	yes	yes	yes	yes	yes	yes	b
User CVs Details									
Vascular Screen Info									
Triggering Details									
Cine Details									
Multi-Phase Details									
Diffusion Screen									
Other Relevant Info:									

Lumbar 8CTL456	SCAN 1	SCAN 2	SCAN 3	SCAN 4	SCAN 5	SCAN 6	SCAN 7	SCAN 8	SCAN 9	SCAN 10
Description	SagT1SE	SagT1FSE	SagPDFSE	SagT2FSE	AxT2FSE	AxT1SE	AxT1FSE	SagT1Flair	AxT2FRFSE	SagT2FRFSE
Coil	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456	8CTL456
Plane	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique	Oblique
Mode	2D	2D	2D	2D	2D	2D	2D	2D	2D	2D
Pulse Sequence	SE	FSE-XL	FSE-XL	FSE-XL	FSE-XL	SE	FSE-XL	T1Flair	FRFSE-XL	FRFSE-XL
Imaging Options	VB	TRF	NPW TRF	NPW	FC NPW	VB	TRF VB	NPW SEQ VB TRF ZIP512	FC NPW TRF ZIP512 VB	NPW TRF ZIP512 VB
PSD Filename										
No. Echos/Shots	1	1	1	1	1	1	1	1	1	1
TE 1/TE 2	MinFull	MinFull	MinFull	102	102	Minimum	Min Full	24	110	110
TR	450	500	2000	3500	4950	600	500	2000	4350	3500
Inv. Time								860		
Flip Angle										
Echo Train Length		2	6	16	12		2	6	15	29
RBW 1/ RBW 2	15.63	25	20.83	20.83	20.83	15.63	20.83	20.83	20.83	41.67
FOV	28	28	28	28	20	20	20	28	20	28
Slice Thickness	4	4	4	4	4	4	4	4	4	4
Slice Spacing	1	1	1	1	1	1	1	1	1	1
No. Scan Locs										
No. Slices/Slabs										
Frequency	256	256	256	256	256	256	256	256	256	448
Phase	224	224	224	224	224	192	192	192	224	224
NEX	3	4	2	4	3	3	3	3	4	4
Phase FOV	0.75	0.75				1	1			
Freq Direction	S/I	S/I	A/P	A/P	A/P	R/L	R/L	A/P	A/P	A/P
Flow Comp Direction										
Autoshim On?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Phase Correct On?	no	no	no	no	no	no	no	no	no	no
No. Locs B4 Pause/Phase										
Sat Details	Anterior	Anterior	Anterior	Anterior	Sup /Inf	Anterior	Anterior	Sup/Inf	Sup/Inf	
SCIC On?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
User CVs Details										
Vascular Screen Info										
Triggering Details										
Cine Details										
Multi-Phase Details										
Diffusion Screen										
Other Relevant Info:										

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2377426-3	550011	1

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup and Test Execution for 11.0 M3A and Greater Software Versions

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position coil so that the top table locators on both sides of the coil fit into the second notch from the magnet. Position the phantom set on the coil without the pad. (see *Figure 7*).

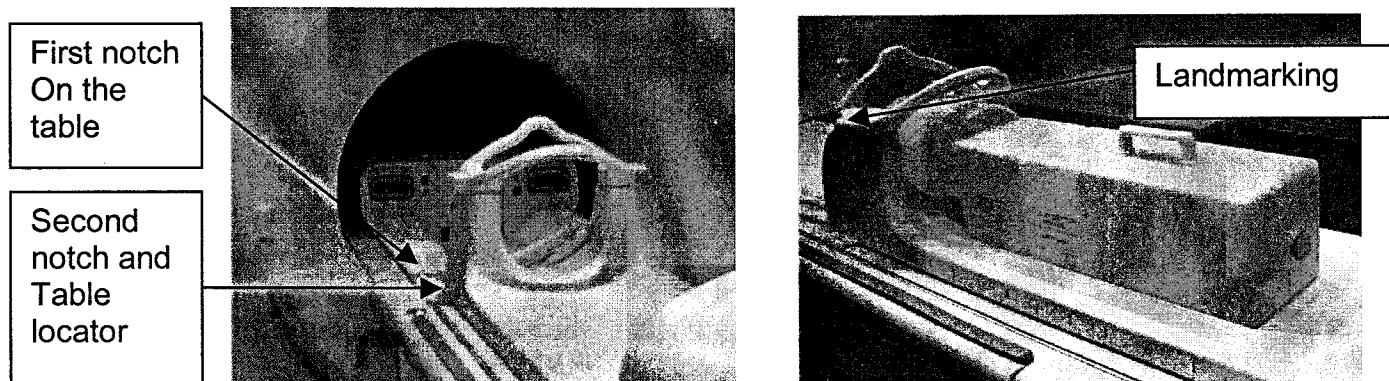


Figure 7: Positioning of the coil and the phantoms on the coil.

3. Landmark the coil at the indicated line on the C-Spine Bridge and advance to scan.
4. From the Common Service Desktop, select **Image Quality** and **Coil SNR**, then [Click here to start this tool.](#)

5. Select the coil type you want to run, enter the serial number and select Start Scan. Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.
6. Follow the reference example below as a visual aid.

- A.  From the Common Service Desktop Manager, select the **Image Quality Tab**.

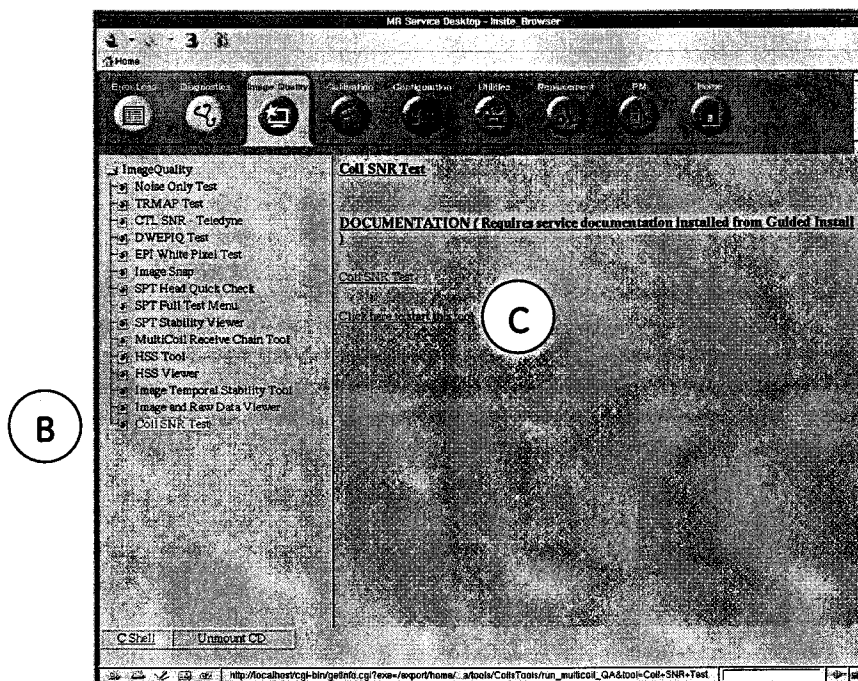


Figure 21- Service Desktop

- B. Select **Coil SNR Test**.
- C. Select **Click here to start this tool**. The Multicoil SNR Tool Control displays.

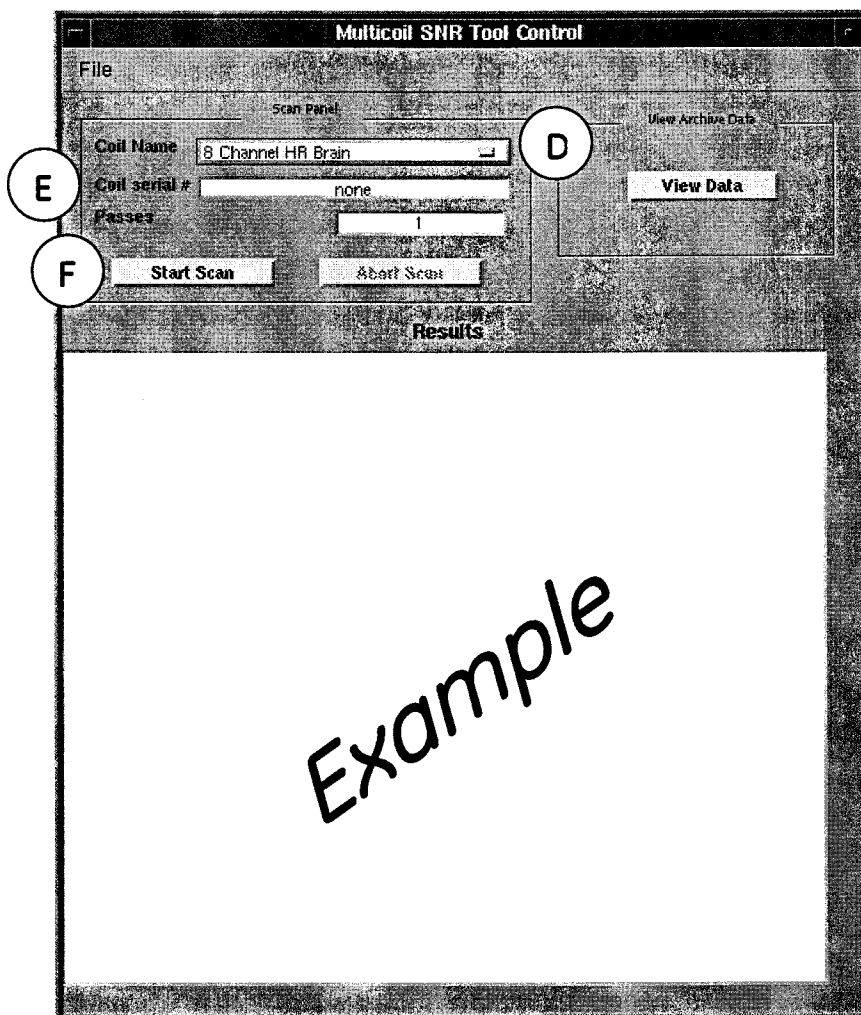


Figure 22 - Multicoil SNR Tool Control

- D. The name of the coil should populate the **Coil Name** field. If desired, select the coil you are testing from the dropdown list.
- E. Enter the **Coil Serial #** found on the coil.
- F. Click **Start Scan** to begin the QA scan.

Note: The scan may take several minutes to complete. For example, an 8-channel coil may take approximately 8 to 10 minutes to complete.

- G. The results of the scan display in the Results window with a Passed or Failed indication (refer to Figure 3).

SNR Data Sheet

Use the table provided below to record the pass/fail results of the SNR test obtained from the Functional Checks section.

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

Date	Serial Number	Pass/Fail	Exam Number

Comments (pass/fail ROI #s)	

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



Do not spray or pour the cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning the coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

4-3 Replaceable Accessories

The table below lists coil accessories that may be purchased for replacement.

REPLACEABLE ACCESSORIES LIST – TABLE 4-3

Part Name	GE Part #	USAI Part #
Coil with Coil ID Cable	2377426-2	165140
Cable Assembly	2377426-7	155131
Adapter Block	2225545-9	155137
Anterior Bridge	2377426-8	155132
Phantom Kit	2377426-3	155135
Patient Comfort Pad Kit	2377426-6	155133

For additional information, contact:

GE Healthcare Americas

USA..... 800-558-5102

Canada 800-668-0732

GE Healthcare Asia

China..... 86-21-62192228

Taiwan 886-2-2505-7900

Singapore..... 65-291-8528

Australia 61-2-9975-5501

Japan 81-120-48-2630

Korea 82-31-740-6119

India 91-80-845-2923

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REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	09/03	Teresa DeMarco	First Issue
2	10/03	Teresa DeMarco	Revised Section 3-2-3, Added Section 1-6 and Translation
3	01/07	Clifton Mills	Revised Section 3-2-3, Revised Serial no. Label